Joint Classification of Hyperspectral Image and LiDAR Data Based on Spectral Prompt Tuning

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***Abstract*— The pretrained vision-language models (VLMs) have achieved outstanding performance in various visual tasks, primarily due to the knowledge they have acquired from massive image–text pairs. This enables VLMs to generalize to a wide range of downstream tasks. This article presents the first attempt to adapt VLMs for the joint classification task of hyperspectral image (HSI) and LiDAR data, aiming to leverage the well-learned VLMs to extract more generalizable features from diverse remote sensing image sources. Initially, using a patch encoder (PE), low-dimensional patches of HSI and LiDAR data are transformed into high-dimensional latent feature representations, meeting the dimensional requirements of VLMs for visual input data. Unlike traditional classifiers that rely on discrete class labels, VLM-based classification methods depend on continuous vectors, which can be derived from textual templates with class names, i.e., prompts. The classification performance of VLM-based methods heavily relies on these prompts, but prompt engineering not only demands extensive expert knowledge but also is extremely time-consuming. To address this, prompt tuning (PT) methods are introduced to enhance the generalizability of VLMs by adding spectral-based prompts to the vision encoder and incorporating randomly initialized, learnable text prompts (TPs) into the text encoder. Finally, through a novel class-discriminative loss function, the distance between text features of different classes is increased, thereby enhancing the model's discriminative ability. Experimental results on the Houston 2013, Trento, and MUUFL datasets demonstrate that the proposed method can achieve competitive classification accuracy with a limited number of labeled pixels.**

***Index Terms*— Classification, hyperspectral image (HSI), LiDAR, prompt tuning (PT), vision-language model (VLM).**

I. Introduction

With the continuous development of remote sensing technology, remote sensing images have been increasingly applied in numerous fields such as surface observation, urban monitoring, agriculture, and military [1], [2]. Among these applications, pixelwise classification stands as a fundamental and significant task [3], [4]. Multimodal remote sensing images can reflect different characteristics of ground objects from various perspectives. For instance, hyperspectral images (HSIs) can reveal the spectral features of ground objects, thereby enabling the identification of subtle differences between various classes of ground objects [5], [6]. LiDAR data, on the other hand, can provide information about the elevation of ground objects [7], [8]. The integrated use of both the modalities of remote sensing images can assist in achieving higher classification accuracy.

Early methods in remote sensing typically comprised two independent phases: feature extraction and subsequent classification [9]. For the feature extraction phase, approaches such as extended morphological profiles (EMPs) and attribute profiles (APs) were frequently used [10], [11]. The fusion process generally depends on a feature stacking mechanism. Nonetheless, these methods often encounter specific challenges: first, handcrafted feature extraction techniques necessitate substantial expert knowledge and are predominantly confined to one or a limited range of feature types; second, the reliance on stacking for feature fusion frequently precipitates the "curse of dimensionality" dilemma. Following the multimodal feature fusion, classification is typically conducted using support vector machines (SVMs) or kernel-based classifiers, to derive the final classification results [12].

Deep learning (DL), renowned for its potent nonlinear mapping capabilities and advanced proficiency in extracting high-level abstract features, has seen the extensive application in joint classification tasks of HSI and LiDAR data [13], [14]. These methods seamlessly integrate the representation learning with classifier learning, using the backpropagation and gradient descent algorithms to facilitate an end-to-end training process. Early deep fusion methods predominantly used convolutional neural networks (CNNs). For instance, Roy et al. [15] introduced a novel approach for the joint classification of HSI and LiDAR data, using a CNN with spatial morphological blocks to extract the spectral–spatial feature from HSI and the geometric information from LiDAR. Li et al. [16] proposed a cross-layer interaction module and a center spectrum correction module to more effectively integrate HSI and LiDAR features. Zhang et al. [17] developed an interleaving perception CNN, which preserves the information through multisource structural constraints and Gram matrix, incorporating the data-driven analysis and task-specific adaptation for efficient parsing of multisource information. Gao et al. [18] extracted the spectral features using an involution operator and integrated them with LiDAR data using a multiscale hierarchical fusion module. This approach

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was further enhanced by an adaptive feature fusion module, optimizing the classifier by dynamically assigning weights to different fusion features based on their contributions. Finally, Wu et al. [19] introduced a cross-channel reconstruction module in CNNs to learn denser fusion features of multisource remote sensing images, achieving the comprehensive interaction between features from different sources. Zhao et al. [20] proposed a novel classification method named factional Gabor convolutional network (FGCN). They used Octave convolution and fractional Gabor convolutional to achieve the interaction of multisource features and the extraction of multiscale, multidirectional, and local semantic variation features, respectively. Besides, the fractional Gabor features and the spectral features were combined with two learnable weighting factors.

To suppress the irrelevant knowledge and enhance the model's focus on significant features, attention mechanism has been incorporated into CNNs. For instance, Wang et al. [21] used a nearest-neighbor-based data augmentation within a contrastive learning framework to improve the semantic alignment between different modalities. In addition, a bilinear attention fusion module is used to effectively capture the complex interactions between HSI and LiDAR features. Song et al. [22] used a two-stream attention-based network for feature extraction and implemented an attentional feature fusion module for both cross-source and single-source data. This method is notable for incorporating a hashing-based similarity metric and a decision-level fusion strategy. Xue et al. [23] leveraged a hierarchical residual structure with self-calibrated convolution to extract multiscale features. They also introduce a spectral and spatial self-attention model for adaptive feature calibration and an attention-based feature fusion mechanism for dynamic and adaptive fusion of features from different modalities. Xu et al. [24] adopted a dynamic multimodal gradient optimization strategy for the adaptive control of gradient modulation across multiple branch networks. Furthermore, to ensure the feature consistency in cross-modal remote sensing images, a multimodal bidirectional enhancement module and a feature distribution consistency loss are devised. CNN-based methods typically use fixed-size patches as network inputs and apply fixed-size convolutional kernels, which makes it challenging to extract features adaptable to different types of land cover. To address this, Feng et al. [25] adopted a dynamic scale feature extraction and hierarchical structures to effectively extract multiscale features and incorporate spatial and modal attention modules for feature fusion. Li et al. [26] implemented the spatial, spectral, and multiscale attention mechanisms to extract the spectral, spatial, and multiscale features from HSI and LiDAR data, respectively, and further used a combined attention to achieve full interaction between features from the two sources.

More recently, an attention-based deep model, known as the transformer, has been increasingly recognized among scholars in the field of remote sensing. For example, Zhao et al. [27] proposed a novel dual-branch method, which combines a hierarchical CNN with a transformer, for effective fusion of multisource heterogeneous information. Dong et al. [28] exploited the hierarchical features of LiDAR and HSI through a multibranch feature fusion architecture. In their works, the self- and cross-guided attention are introduced at each layer of the network to emphasize the weights of regions and channels of interest, while a spectral supplementary module is used to enhance the discriminative capability of the central pixel, thus enabling effective feature extraction and fusion between LiDAR and HSI features. Yu et al. [29] proposed an HSI intrinsic decomposition-guided data fusion network that integrates the horizontal geometric information of HSI shadow component with the vertical geometric data of LiDARs. Zhang et al. [30] implemented a spectral transformer and a spatial transformer to extract the spectral and spatial features of HSI and LiDAR data, respectively, and implemented a fusion transformer for the interaction and fusion of features from different sources. Ding et al. [31] merged the local spatial feature extraction capability of CNN with the advantage of transformer in learning global spectral context information, effectively enhancing the performance of multimodal remote sensing image fusion and classification. Yao et al. [32] introduced a cross-modal attention mechanism that mines pixel-level spatial similarities, promoting the feature fusion between multimodal features. Zhao et al. [33] proposed an efficient fractional Fourier image transformer to effectively extract the global and local features of multimodal remote sensing images. Roy et al. [35] proposed a multihead cross-patch attention-based transformer to use the complementary features of multisource data. With the help of their method, feature interactions between different modalities are effectively promoted.

Both the CNN and transformer-based methods typically require a large number of labeled samples for training. Otherwise, they may encounter serious overfitting issues, potentially diminishing the accuracy on testing samples. To address this challenge, numerous semi-supervised learning theories have been applied to effectively leverage limited labeled samples along with a large pool of unlabeled samples. For instance, Li et al. [34] used three CNN branches to extract the spectral, spatial, and elevation features, integrating the feature-level and decision-level fusion strategies. They introduced a unique label calibration module to generate effective pseudolabels for semi-supervised learning. In addition, a 3-D cross attention block is designed to enhance the complementary spatial features of multisource data. Xiu et al. [36] used a semi-supervised graph transductive module to fully explore the correlations between pixels from scarce labeled samples and an abundance of unlabeled samples. Lin et al. [37] proposed a masked autoencoder network with dual branches for the spatialwise and spectralwise analysis, which uses an unsupervised approach involving random patches and band reconstruction to extract features from different sources.

Different regions of the same ground objects may exhibit either homogeneous or heterogeneous features, providing a foundation for contrastive learning strategies that focus on positive–negative sample pairs. Several DL methods based on contrastive learning strategies have been proposed and applied to HSI and LiDAR data classification tasks. These methods are designed to automatically learn general features of multimodal data in scenarios with limited labeling. For instance, Jia et al. [38] introduced a collaborative contrastive learning approach

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for HSI and LiDAR data classification. This method involves two stages: pretraining for collaborative feature representation and fine-tuning for multilevel remote sensing feature fusion. Feng et al. [39] developed the cross-modal and intramodal contrastive learning strategies to ensure semantic consistency within different modalities of remote sensing images in the same area. They implemented a mixed cross-modal fusion module to effectively integrate the complementary information from multiple modalities.

Almost all the existing DL-based methods need the training of a well-designed deep network from scratch. However, in the fields of computer vision, numerous pretrained DL models are easily accessible. Particularly in recent years, many large-scale pretrained vision-language models (VLMs) trained on massive image–text pairs have been introduced, such as CLIP [40], ALIGN [41], and others. This raises an intriguing question: can these pretrained VLMs be effectively generalized to the joint classification task of HSI and LiDAR? Intuitively, while remote sensing images differ from conventional natural images, they also share similar features (such as edges and contours). To answer this question, inspired by the concept of prompt tuning (PT) methods, this article proposes a spectral PT (SPT)-based method for the joint classification of HSI and LiDAR, leveraging a pretrained VLM named CLIP [40]. The main contributions of this article are outlined below.

1) This article attempts to generalize VLMs to the joint classification task of HSI and LiDAR data. To achieve this objective, while maintaining the parameters of the inherent vision encoder and text encoder of the standard CLIP model fixed, three additional learnable components are introduced: a patch encoder (PE), a spectral prompt learner (SPL), and text prompts (TPs).
2) To use the abundant spectral features of HSI and to avoid the problem of overfitting to downstream tasks, a learnable SPL is constructed. The sample-dependent spectral prompt vectors (SPVs) obtained from SPL are input into the vision encoder along with frozen visual tokens. This enables the output features to encompass the spatial, spectral, and elevation information.
3) Considering that only using class names for text embedding may inadequately describe the entire class, SPT enhances original text embedding using the learnable class TP, thereby enriching the description of classes. In addition, a class similarity matrix is used to characterize the similarity between text embedding vectors of different classes. By minimizing the elements on nondiagonal lines, the similarity between different classes is reduced, thereby enhancing the discriminative ability of the classification model.

II. SPT-Based Joint Classification of HSI and LiDAR Data

As shown in Fig. 1, SPT-based joint classification of HSI and LiDAR data method mainly contains five steps.

1) Small patches are extracted from the original HSI and LiDAR data by selecting several pixels around a specific pixel, and then concatenating the features of both the modalities along the channel dimension. Subsequently, a corresponding high-dimensional feature representation is obtained through a learnable PE.
2) Obtain the spectral prompts (SPs) by an SPL.
3) Multiple frozen visual tokens are obtained through an embed model, and these tokens are then input into the transformer encoder layers of CLIP, along with the class (CLS) token and learnable SPs, to yield the image features.
4) For the text part of CLIP, the class name (represented by [CLASS]) along with learnable TPs are input into the text encoder to obtain the corresponding text features.
5) Based on the text and image features, a classification loss (L_{CE}) is calculated. Moreover, the similarity between different classes is computed based on text features, which further leads to the calculation of a class discriminative (CD) loss (L_{CD}).

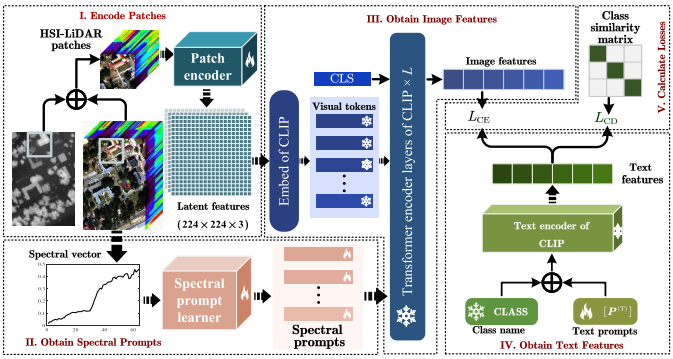


Fig. 1. Flowchart of SPT for joint classification of HSI and LiDAR.

A. Patch Encoder

Given the original HSI $\boldsymbol{X}^{(\mathrm{HSI})} \in \mathbb{R}^{H\times W\times D}$ and LiDAR data $\boldsymbol{X}^{(\mathrm{LiDAR})} \in \mathbb{R}^{H\times W\times d}$. H is the height of HSI and LiDAR data, W is the width of HSI and LiDAR data, D and d are the numbers of channels of HSI and LiDAR data, respectively. Small patches of HSI and LiDAR data $\tilde{\boldsymbol{X}}^{(\mathrm{HSI})} \in \mathbb{R}^{h\times h\times D}$ and $\tilde{\boldsymbol{X}}^{(\mathrm{LiDAR})} \in \mathbb{R}^{h\times h\times d}$ can be obtained by extracting several neighboring pixels around a specific pixel of corresponding source data, respectively. h represents the width and height of small patches of both HSI and LiDAR data. Typically, the value of h is relatively small (e.g., 11) with a high number of channels, rendering the input data unable to meet the dimensional requirements of image encoder in CLIP. To address this issue, a multilayer PE is constructed primarily using the transposed convolution, which increases the spatial dimensions and decreases the spectral dimensions

$$\boldsymbol{X}' = f_{\boldsymbol{\theta}^{(\mathrm{PE})}}\left(\tilde{\boldsymbol{X}}^{(\mathrm{HSI})} \oplus \tilde{\boldsymbol{X}}^{(\mathrm{LiDAR})}\right) \tag{1}$$

where $\oplus$ means the operation of concatenation. $\boldsymbol{X}' \in \mathbb{R}^{224\times 224\times 3}$, which is determined by the selected pretrained VLMs, and $\boldsymbol{\theta}^{(\mathrm{PE})}$ represents the parameters of PE. PE consists of multiple stages of transposed convolution blocks, each comprising a transposed convolution layer, a batch normalization (BN) layer, and a nonlinear function.

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B. SP Learner

The most straightforward method to adapt a pretrained VLM to the joint classification of HSI and LiDAR data is to use fine-tuning techniques, i.e., fine-tune the entire model's parameters on the HSI and LiDAR datasets. However, this requires updating millions or even billions of parameters, which not only demands a vast amount of labeled data for support but also consumes considerable time and computational resources, and could even significantly disrupt the well-learned model. Therefore, this study considers the introduction of PT method [42].

CLIP typically consists of two components: a vision encoder and a text encoder. For the vision encoder, the input image goes through feature extraction by multiple visual transformer blocks, further yielding high-level features. For the text encoder part, the class-related text like "a small region of [CLASS]" is input and features of the same dimension as the vision encoder can be obtained. The "[CLASS]" can be replaced by specific class names, such as "healthy grass" and "water." Naturally, CLIP-based PT can be considered from both text and vision perspectives. For the vision branch, inspired by visual PT (VPT) [43], we first divide the high-dimensional features $\boldsymbol{X}'$ output by PE into a series of small patches $\boldsymbol{x}_j$ (in this article, the dimension of each small patch is chosen to be 32×32). Then, each small patch containing local spatial information of latent features will be transformed via embedding to obtain $\boldsymbol{e}_0^j$

$$\boldsymbol{e}_0^j = \text{Embed}(\boldsymbol{x}_j). \tag{2}$$

If the high-dimensional feature vectors output by PE are directly input into the vision encoder of CLIP for feature extraction, the obtained features may not adequately reflect the subtle difference between classes of ground objects. This is because, on one hand, the high-dimensional feature representation input into the vision encoder significantly differs from natural images used in CLIP's pretraining; on the other hand, the feature extraction procedure of vision encoder is spatial-oriented, and the relationships between different small patches are explored by the self-attention mechanism. Only spatial-oriented feature extraction is insufficient for accurately distinguishing complex ground objects, as there are always some heterogeneous pixels with similar spatial or elevational features. Fortunately, in HSI, each pixel corresponds to a dense and approximately continuous spectral vector. Therefore, this article designs a sample-dependent SPT method to fully use the abundant spectral information. Given the spectral vector $\boldsymbol{v}_i$ corresponding to each pixel, the SPV can be learned by an SPL, which is constructed using a multilayer perceptron (MLP) or a 1D-CNN. For the convenience in generating multiple SPVs, this article chooses a 1D-CNN. Then, the calculation method for SPVs corresponding to each pixel is

$$\boldsymbol{P}_i^{(\text{S})} = f_{\boldsymbol{\theta}^{(\text{SPL})}}(\boldsymbol{v}_i) \tag{3}$$

where $\boldsymbol{\theta}^{(\text{SPL})}$ represents the learnable parameters of SPL. It can be seen that the obtained SPVs can reflect the spectral features of each pixel. Then SPVs and $\boldsymbol{e}_0^j$ are fed into the transformer layer of the vision encoder

$$\begin{cases} \boldsymbol{E}^l = g^l\left(\left[\boldsymbol{E}^{l-1}, \boldsymbol{P}^{(\text{S})}\right]\right), & \text{if } l = 1 \\ \boldsymbol{E}^l = g^l\left(\left[\boldsymbol{E}^{l-1}\right]\right), & \text{if } l > 1 \end{cases} \tag{4}$$

where $g^l(\cdot)$, $l = 1, 2, \ldots, L$, represents the lth transformer layer, L is the total number of transformer layer, and $[\cdot, \cdot]$ represents the concatenation operator along the sequence dimension. Equation (4) indicates that SPVs are only added into the first transformer layer. Thus, the overall image encoder $G_{\boldsymbol{\theta}_{\text{SPL}}}(\cdot)$ can be represented as

$$\boldsymbol{I}^{(\text{IM})} = G_{\boldsymbol{\theta}_{\text{SPL}}}\left(\left[\boldsymbol{E}^{l-1}, \boldsymbol{P}^{(\text{S})}\right]\right) \tag{5}$$

where $\boldsymbol{I}^{(\text{IM})}$ is the image features. Typically, the obtained image features are subjected to a normalization process $\tilde{\boldsymbol{I}}^{(\text{IM})} = \boldsymbol{I}^{(\text{IM})}/\|\boldsymbol{I}^{(\text{IM})}\|$, where $\|\cdot\|$ denotes the norm operation. By adding the SPL block, the information interaction between spatial-oriented tokens and spectral-oriented tokens can be realized. Subsequently, the transformer of CLIP's vision encoder will aggregate more corresponding features into the CLS token. In this way, SPT can extract spectral–spatial-elevation fusion features with different modal preferences according to the needs of the downstream classification task, thus helping to obtain more accurate classification results.

C. Text PT

Unlike conventional DL-based methods, CLIP first computes the cosine similarity between visual features and various text features. Then, the class corresponding to text feature with the highest similarity is selected as the predicted class. This approach leverages the capability of CLIP to align and compares features across modalities (visual and textual), allowing for more nuanced and context-aware classification. Specifically, a manually constructed prompt like "a small region of [CLASS]" is first converted into a byte pair encoding representation [44]. Each word corresponds to a unique numeric ID. Then, each text sequence is combined with the [SOS] and [EOS] to form a representation with fixed length of 77. Furthermore, the representation will be mapped to 512-D word embedding vectors and passed onto the text encoder of CLIP. However, hand-crafted prompts may inadequately describe the entire class. Samples with the same class may be associated with various semantic descriptions. Therefore, some meaningful contexts should be added to the manually constructed prompts to obtain more reasonable descriptions for classes.

TPs commonly include hard prompts and soft ones. For the former, prompts with explicit semantic meanings are used to construct specific templates. However, hard prompts usually require significant manual effort to design specific prompt templates for different downstream tasks. Therefore, more flexible soft prompts are used in this study. Different from hard prompts, soft prompts use learnable vectors $\boldsymbol{P}^{(\text{TE})}$ to enhance the word embedding vectors $\boldsymbol{O} = \{\boldsymbol{o}_1, \boldsymbol{o}_2, \ldots, \boldsymbol{o}_C\}$ corresponding to [CLASS], i.e., $[\boldsymbol{P}^{(\text{TE})}, \boldsymbol{o}_c]$, $c = 1, 2, \ldots, C$, where C is the total number of classes. Furthermore, using the text encoder $G_{\boldsymbol{P}^{(\text{TE})}}(\cdot)$ to map the enhanced word embedding

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vectors, the text features corresponding to each class can be obtained

$$\boldsymbol{I}_c^{(\mathrm{TE})} = G_{\boldsymbol{P}^{(\mathrm{TE})}}\left(\left[\boldsymbol{P}^{(\mathrm{TE})}, \boldsymbol{o}_c\right]\right) \tag{6}$$

where $\boldsymbol{I}_c^{(\mathrm{TE})}$ denotes the text features corresponding to class c, and $\boldsymbol{P}^{(\mathrm{TE})}$ is the randomly initialized learnable prompt vector. Similar to the vision part, after obtaining the text features, the normalization is also performed $\tilde{\boldsymbol{I}}^{(\mathrm{TE})} = \boldsymbol{I}^{(\mathrm{TE})}/\|\boldsymbol{I}^{(\mathrm{TE})}\|$.

D. CD Loss

The text features calculated by (6) describe the semantic characteristics corresponding to each class. In addition, these semantic features can also be viewed as weight vectors corresponding to different classes, which means that the probability of each sample belonging to a certain class can be expressed as

$$p\left(c|\boldsymbol{X}'\right) = \frac{\exp\left(\mathrm{Sim}\left(\tilde{\boldsymbol{I}}^{(\mathrm{IM})}, \tilde{\boldsymbol{I}}_c^{(\mathrm{TE})}\right)/\tau\right)}{\sum_{i=1}^{C}\exp\left(\mathrm{Sim}\left(\tilde{\boldsymbol{I}}^{(\mathrm{IM})}, \tilde{\boldsymbol{I}}_i^{(\mathrm{TE})}\right)/\tau\right)} \tag{7}$$

where τ is the temperature coefficient, $\mathrm{Sim}(\cdot,\cdot)$ denotes the similarity calculation function, and $\mathrm{Sim}(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{y}$ is selected here. Then, the classification loss L_{C} can be further expressed as

$$L_{\mathrm{CE}} = -\log\frac{\exp\left(\mathrm{Sim}\left(\tilde{\boldsymbol{I}}^{(\mathrm{IM})}, \tilde{\boldsymbol{I}}_c^{(\mathrm{TE})}\right)/\tau\right)}{\sum_{i=1}^{C}\exp\left(\mathrm{Sim}\left(\tilde{\boldsymbol{I}}^{(\mathrm{IM})}, \tilde{\boldsymbol{I}}_i^{(\mathrm{TE})}\right)/\tau\right)}. \tag{8}$$

Furthermore, if the semantic similarity between classes is strong, the distance between their text features in a latent space will be close, potentially leading to interclass sample mixing. To mitigate this, a learning strategy is adopted to minimize the intraclass semantic similarity, ensuring that the text features of different classes are far away from each other, thereby enhancing the discriminative ability of classification model. To achieve this, the similarity between text features is calculated as follows:

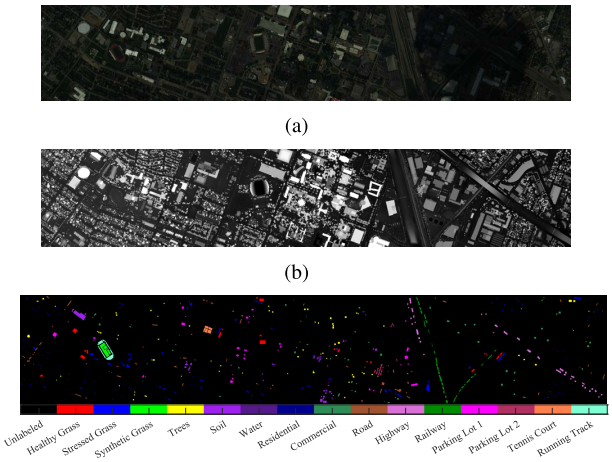
$$\boldsymbol{S} = \mathrm{Sim}\left(\tilde{\boldsymbol{I}}^{(\mathrm{TE})}, \tilde{\boldsymbol{I}}^{(\mathrm{TE})}\right) \tag{9}$$

where $\boldsymbol{S} \in \mathbb{R}^{C\times C}$, and the elements on the diagonal represent the similarity of each class with itself. Since the normalization is performed, the elements on the diagonal of $\boldsymbol{S}$ are always 1. The off-diagonal elements represent the similarity between classes. To reduce the similarity between text features of different classes and thereby increase the distinguishability between samples of different classes, it is necessary to minimize the values on the off-diagonal. In the extreme case, where the elements on the diagonal of the class similarity matrix are 1 and the off-diagonal elements are 0, we use $\boldsymbol{I}_C$ to represent this matrix. This extreme case can be considered as the ultimate learning goal of matrix $\boldsymbol{S}$. Therefore, the following learning objective can be constructed:

$$L_{\mathrm{CD}} = \sigma(\boldsymbol{S}, \boldsymbol{I}_C) \tag{10}$$

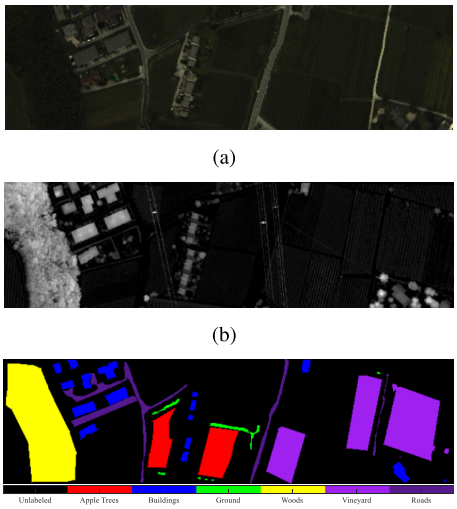
where $\sigma(\cdot,\cdot)$ denotes the mean squared error function. In summary, the overall learning objective function for SPT is

$$L = L_{\mathrm{CE}} + \lambda L_{\mathrm{CD}} \tag{11}$$



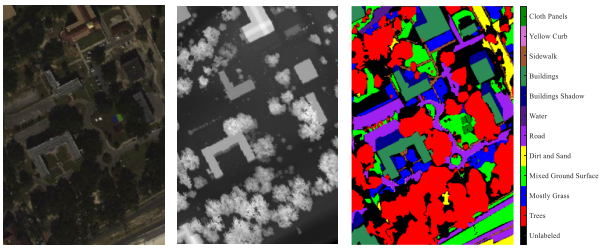
(a) (b) (c)

Fig. 2. Houston 2013 dataset. (a) Pseudo-color composite image. (b) Grayscale image for LiDAR-based DSM. (c) Ground-truth map.



(a) (b) (c)

Fig. 3. Trento dataset. (a) Pseudo-color composite image. (b) Grayscale image for LiDAR-based DSM. (c) Ground-truth map.



(a) (b) (c)

Fig. 4. MUUFL dataset. (a) Pseudo-color composite image. (b) Grayscale image for LiDAR-based DSM. (c) Ground-truth map.

where λ is the balancing coefficient. According to (11), in conjunction with an optimization method similar to AdamW, iterative updates can be performed on learnable parameters $\boldsymbol{\theta}^{(\mathrm{PE})}$, $\boldsymbol{\theta}^{(\mathrm{SPL})}$, and $\boldsymbol{P}^{(\mathrm{TE})}$.

III. Experiments and Analysis

To validate the effectiveness of the proposed method, experiments and analysis were conducted on three real remote

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TABLE I
SAMPLE INFORMATION FOR THE THREE DATASETS

	Houston 2013			Trento			MUUFL		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
C1	Healthy Grass	20	1231	Apple Trees	20	4014	Trees	20	23226
C2	Stressed Grass	20	1234	Buildings	20	2883	Mostly Grass	20	4250
C3	Synthetic Grass	20	677	Ground	20	459	Mixed Ground Surface	20	6862
C4	Trees	20	1224	Woods	20	9103	Dirt and Sand	20	1806
C5	Soil	20	1222	Vineyard	20	10481	Road	20	6667
C6	Water	20	305	Roads	20	3154	Water	20	446
C7	Residential	20	1248				Buildings Shadow	20	2213
C8	Commercial	20	1224				Buildings	20	6220
C9	Road	20	1232				Sidewalk	20	1365
C10	Highway	20	1207				Yellow Curb	20	163
C11	Railway	20	1215				Cloth Panels	20	249
C12	Parking Lot 1	20	1213						
C13	Parking Lot 2	20	449						
C14	Tennis Court	20	408						
C15	Running Track	20	640						
	Total	300	14729		120	30094		220	53467

TABLE II
STRUCTURE OF PE

Block type	Kernel number @ size	Step size /padding size	Output size
Input	-	-	11×11×(11/12)
2D Transp. Conv.	32 @ 4×4	3 / 3	28×28×32
2D Transp. Conv.	32 @ 2×2	2 / 0	56×56×32
2D Transp. Conv.	32 @ 2×2	2 / 0	112×112×32
2D Transp. Conv.	3 @ 2×2	2 / 0	224×224×3

sensing datasets: Houston 2013, Trento, and MUUFL. Each of these datasets consists of an HSI and LiDAR data. The Houston 2013 dataset was provided by the 2013 IEEE GRSS Data Fusion Contest [45]. It consists of 144 spectral bands in the HSI and one channel in the LiDAR data, with a total of 15 classes. The Trento dataset [46] was acquired using the AISA Eagle system and comprises six distinct land cover classes. It includes 63 spectral bands in the HSI and one channel in the LiDAR data. The MUUFL dataset [47] was captured in 2010 using a reflective optics system imaging spectrometer sensor. It encompasses a total of 11 classes. The HSI includes 64 spectral bands, and the LiDAR data comprise two raster channels. For more detailed sample information, false-color composites of the HSI, grayscale representations of the LiDAR data, and the ground-truth labels, can be found in Table I and Figs. 2–4.

A. Comparative Experiments

To validate the effectiveness of the proposed method, the following methods are chosen for comparative experiments, including SVM, Coupled-CNN [4], DSHFNet [25], NNCNet [21], HCT [27], ExViT [32], FGCN [20], and MFT [35]. The hyperparameters of SVM are selected through cross-validation, while the hyperparameters of other comparative methods are set according to the respective reference. To reduce the randomness introduced by the selection of training samples, all the experiments are repeated five times, and the statistical averages are computed. All the experiments are performed on a computer equipped with 3.70-GHz Intel Core I9-10900KF CPU 32 GB of RAM, RTX 4090 GPU, and the experimental environment is PyTorch. The detailed network structures of PE and SPL in SPT are given in Tables II and III. The PE includes four transpose convolutional blocks, each comprising a transposed convolution layer, a BN layer, and a nonlinear activation function (GELU) [48]. The band of HSI is first reduced to ten with principal component analysis (PCA) to reduce the computational cost. Then, the constructed HSI and LiDAR patches are concatenated along the channel dimension. Therefore, the size of patches input into PE is $11 \times 11 \times 11$ for the Houston and Trento datasets, and $11 \times 11 \times 12$ for the MUUFL dataset. The dimension of the output from the PE network is $224 \times 224 \times 3$, meeting the requirements of the CLIP vision encoder for input dimensions. The ViT-32/B is selected as our base model, which is the "Base" variant with 32×32 input patch size.

SPT primarily consists a 1-D convolutional block, each block comprising a 1-D convolutional layer, a layer normalization, and a nonlinear activation function, with slight variations in settings across different datasets. The kernel sizes on the three datasets are 36, 21, and 16, respectively, each accompanied by 768 nonoverlapping 1-D convolutional kernels. As a result, the dimensions of the output are 768×4, 768×3, and 768×4, which means that on the three datasets, the number of added SPVs is 4, 3, and 4, respectively.

In addition, on Houston 2013, the learning rate is set to 0.001, while on the other two datasets, it is set to 0.005. On Houston 2013 and Trento, the number of iterations is set to 200, while on MUUFL, the number of iterations is set to 100. The batch size is set 32 on all the three datasets, and AdamW optimizer is selected. On Trento and MUUFL, λ is set to 0.01, and on Houston 2013, it is set to 0.02. The comparison of classification performance on different datasets is shown in Tables IV–VI, and Figs. 5–9. For better observation, local regions are enlarged and shown in Figs. 6 and 8. The t-distributed stochastic neighbor embedding (t-SNE) [49] visualization of features extracted by SPT is shown in Fig. 10. The following can be seen.

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TABLE III
STRUCTURE OF SPL

	Input size	Layer type	Kernel number @ size	Step size / padding size	Output size
Houston 2013	144	1D Conv.	768 @ 36	36 / 0	768×4
Trento	63	1D Conv.	768 @ 21	21 / 0	768×3
MUUFL	64	1D Conv.	768 @ 16	16 / 0	768×4

TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCE (HOUSTON 2013 DATASET)

	SVM	Coupled-CNN [4]	DSHFNet [25]	NNCNet [21]	HCT [27]	ExViT [32]	FGCN [20]	MFT [35]	SPT
Healthy Grass (%)	**95.65**	95.34	92.06	91.62	92.95	91.37	95.53	94.02	94.25
Stressed Grass (%)	94.57	93.24	94.83	95.12	94.29	95.19	89.79	92.66	**95.90**
Synthetic Grass (%)	99.11	99.44	**100**	99.65	99.38	99.05	98.97	99.65	99.94
Trees (%)	93.53	97.96	**98.45**	97.39	97.24	97.53	97.63	96.62	98.33
Soil (%)	94.81	99.97	99.35	99.89	**99.98**	99.69	98.61	99.18	99.82
Water (%)	90.43	**97.05**	95.48	96.98	96.85	94.56	88.52	96.00	96.39
Residential (%)	84.62	91.97	95.05	91.83	92.80	94.17	91.19	91.06	**96.22**
Commercial (%)	75.36	82.35	82.65	**85.85**	81.13	78.91	80.64	76.63	76.88
Road (%)	73.69	75.76	59.29	80.76	76.92	85.19	80.76	84.01	**87.91**
Highway (%)	79.59	88.68	89.54	88.96	90.80	87.79	91.38	**91.40**	86.18
Railway (%)	78.77	95.69	97.43	**98.95**	98.53	97.23	76.87	89.42	91.37
Parking Lot 1 (%)	77.56	91.41	90.67	88.69	**95.10**	91.84	76.42	91.23	94.33
Parking Lot 2 (%)	45.03	**97.55**	95.77	94.79	97.37	97.24	87.75	96.70	95.32
Tennis Court (%)	98.19	**100**	98.53	99.90	**100**	99.36	97.79	99.56	99.80
Running Track (%)	98.75	**100**	99.47	99.97	**100**	99.78	**100**	99.53	**100**
AA (%)	85.31	93.76	92.57	94.02	**94.22**	93.93	90.12	93.18	94.18
OA (%)	85.37	92.55	91.57	93.02	93.14	92.99	89.22	91.97	**93.23**
Kappa	0.8417	0.9194	0.9062	0.9246	0.9258	0.9242	0.8835	0.9132	**0.9268**
Time (s)	**7.40**	22.18	680.36	4420.32	437.90	86.67	2010.34	1176.70	130.89

TABLE V
COMPARISON OF CLASSIFICATION PERFORMANCE (TRENTO DATASET)

	SVM	Coupled-CNN [4]	DSHFNet [25]	NNCNet [21]	HCT [27]	ExViT [32]	FGCN [20]	MFT [35]	SPT
Apple Trees (%)	83.26	98.75	97.90	96.71	98.85	98.75	97.06	97.14	**99.48**
Buildings (%)	92.66	91.72	95.52	93.88	95.28	94.97	97.50	97.02	**97.73**
Ground (%)	96.17	99.04	97.30	96.56	98.69	**99.13**	96.30	98.00	98.56
Woods (%)	98.00	**100**	99.91	99.04	99.97	**100**	99.99	99.99	99.98
Vineyard (%)	80.00	**100**	99.97	94.66	98.99	99.98	98.52	98.77	99.66
Roads (%)	89.04	95.99	96.08	92.54	95.19	97.01	**97.53**	95.61	95.68
AA (%)	89.86	97.58	97.78	95.56	97.83	98.31	97.82	97.75	**98.51**
OA (%)	88.29	98.60	98.80	95.99	98.51	99.02	98.53	98.41	**99.11**
Kappa	0.8460	0.9814	0.9840	0.9467	0.9801	0.9869	0.9805	0.9788	**0.9881**
Time (s)	**0.90**	13.72	253.63	631.90	42.46	37.63	244.48	641.21	82.60

TABLE VI
COMPARISON OF CLASSIFICATION PERFORMANCE (MUUFL DATASET)

	SVM	Coupled-CNN [4]	DSHFNet [25]	NNCNet [21]	HCT [27]	ExViT [32]	FGCN [20]	MFT [35]	SPT
Trees (%)	78.14	84.84	71.23	79.18	83.49	84.07	84.31	82.50	**84.93**
Mostly Grass (%)	74.66	65.61	67.43	**77.02**	73.05	66.47	58.45	68.42	76.07
Mixed Ground Surface (%)	58.77	67.07	56.93	62.17	67.87	60.10	57.00	60.06	**73.93**
Dirt and Sand (%)	75.87	**93.13**	90.71	86.62	89.62	84.74	76.74	70.97	89.27
Road (%)	86.76	80.29	81.90	79.17	77.97	74.29	68.16	72.53	**89.15**
Water (%)	95.87	99.82	62.29	99.87	**100**	99.46	91.93	97.09	98.48
Buildings Shadow (%)	87.23	89.42	90.20	80.40	83.80	84.23	86.76	80.12	**92.00**
Buildings (%)	84.63	87.04	79.06	91.27	92.93	**95.23**	81.45	89.16	91.21
Sidewalk (%)	58.59	51.94	51.60	50.33	52.63	53.23	70.84	60.70	**76.01**
Yellow Curb (%)	87.12	71.41	67.48	79.26	79.88	67.85	51.53	62.45	**96.32**
Cloth Panels (%)	93.41	98.31	96.39	97.75	97.27	96.71	**98.39**	95.26	95.18
AA (%)	80.10	80.81	74.11	80.28	81.68	78.76	75.05	76.30	**87.50**
OA (%)	77.25	80.49	72.31	78.06	80.69	79.05	75.93	77.11	**84.48**
Kappa	0.7132	0.7512	0.6578	0.7228	0.7542	0.7333	0.6936	0.7099	**0.8016**
Time (s)	**1.76**	16.09	404.42	483.83	61.43	79.36	186.35	901.14	104.04

1) On all the datasets, SPT achieves an OA that is competitive with the latest state-of-the-art methods, demonstrating the feasibility of generalizing CLIP to the joint classification task of HSI and LiDAR data.

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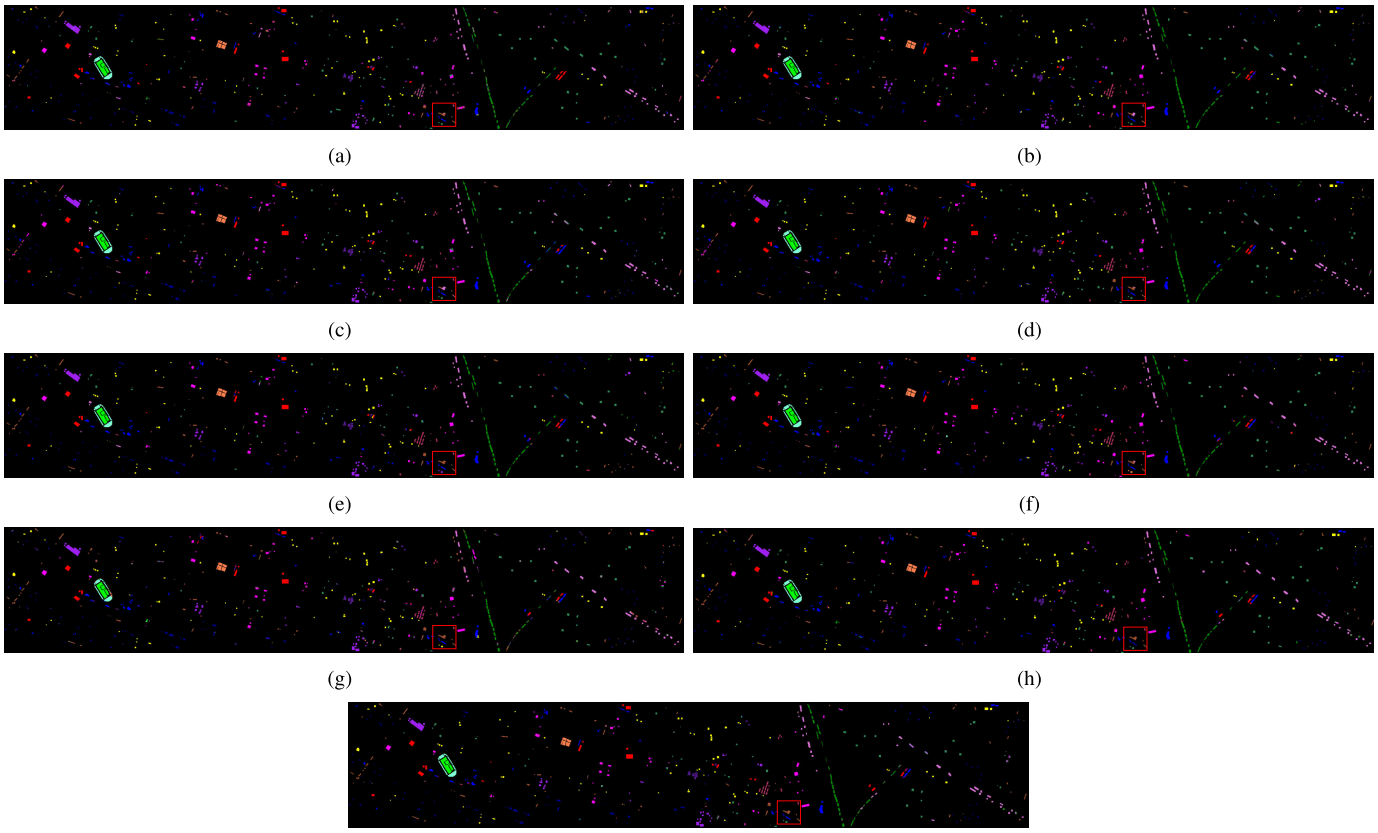


Fig. 5. Classification maps of different methods on the Houston 2013 dataset. (a) SVM. (b) Coupled-CNN. (c) DSHFNet. (d) NNCNet. (e) HCT. (f) ExViT. (g) FGCN. (h) MFT. (i) SPT.

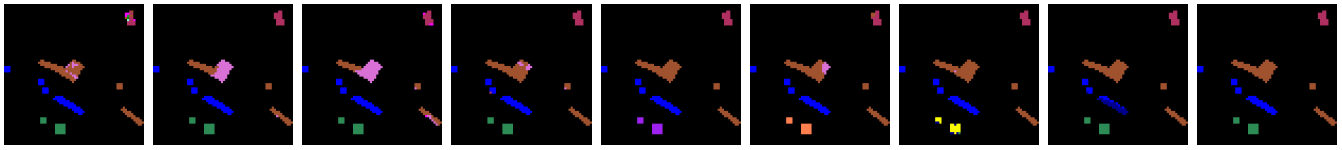


Fig. 6. Classification maps of a local region obtained by different methods on the Houston 2013 dataset. (a) SVM. (b) Coupled-CNN. (c) DSHFNet. (d) NNCNet. (e) HCT. (f) ExViT. (g) FGCN. (h) MFT. (i) SPT.

By adding learnable PE, SPL, and TPs, while keeping the parameters of CLIP frozen, we not only retain CLIP's original powerful visual feature extraction capabilities and strong semantics but also enable its generalization to the joint classification task of HSI and LiDAR data with a limited number of learnable parameters. This approach avoids the risk of network collapse that could result from adjusting all the network parameters.

2) Compared with the Houston 2013 and Trento datasets, all the methods achieve lower classification accuracy on the MUUFL dataset. This is due to the presence of samples with high interclass similarity on the MUUFL dataset. For instance, classes "Road" and "Sidewalk" are mostly spatially adjacent and dispersed, making the classification of these two classes challenging. SPT achieves the highest classification accuracy for these two classes, benefiting from the minimized interclass similarity, which increases the distance between classes in the text feature space.
3) All the DL methods achieve very high OAs on the Houston 2013 and Trento datasets, but they exhibit relatively lower OAs on the MUUFL dataset. This lower performance is attributed to the more complex data distribution and higher interclass similarity on the MUUFL dataset. The limited labeled samples make it challenging to adequately represent the true data distribution, and as a result, DL methods tend to overfit to limited labeled samples, leading to reduced generalizability to test samples. Despite these challenges, SPT still manages to achieve relatively higher OA on the MUUFL dataset.
4) Compared with other methods, SPT can yield classification maps that are smoother, more detailed, and have less salt-and-pepper noise. SPT is able to classify Stressed Grass, Road, and Commercial more accurately in the local region on the Houston 2013 dataset, whereas

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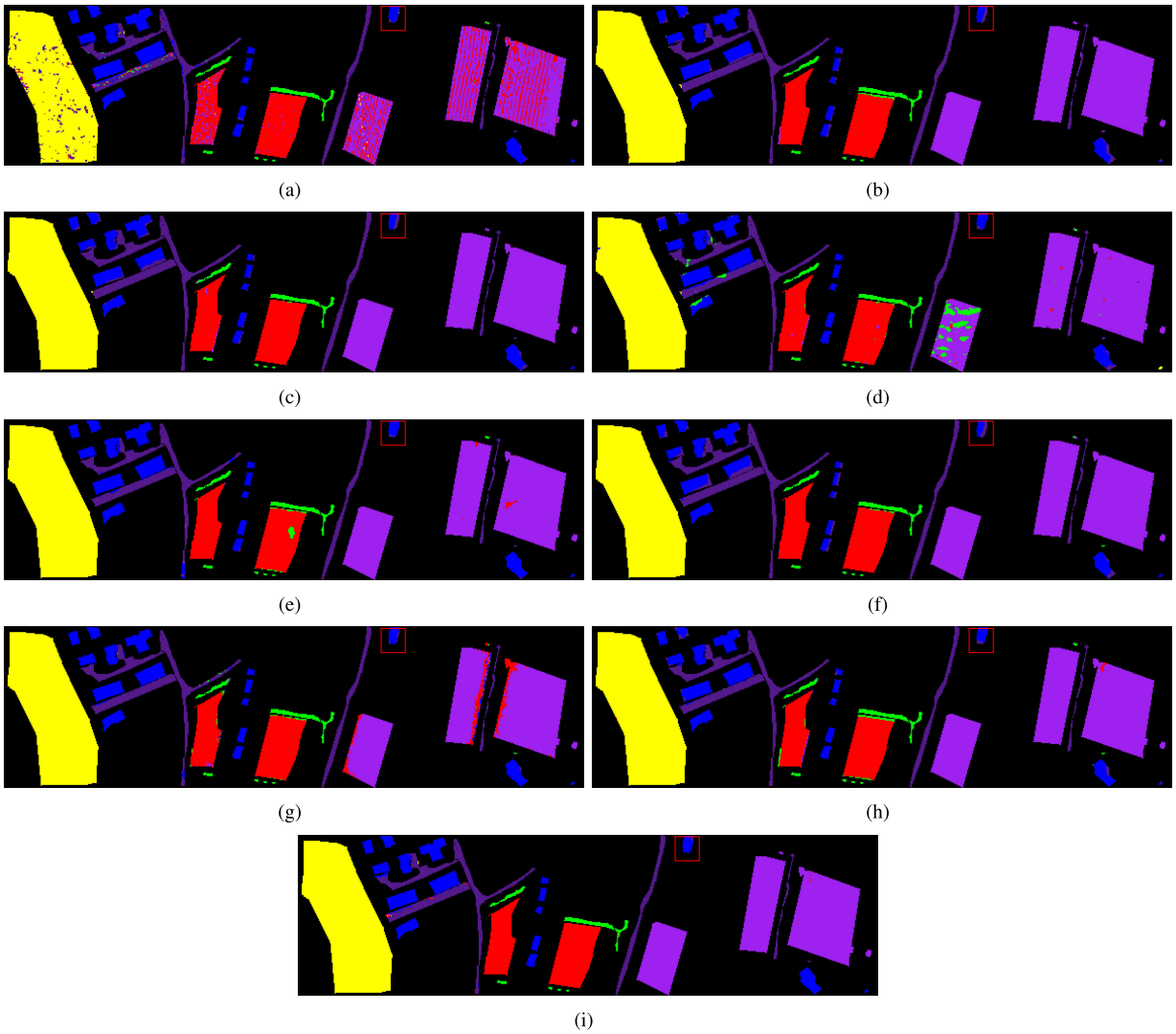


Fig. 7. Classification maps of different methods on the Trento dataset. (a) SVM. (b) Coupled-CNN. (c) DSHFNet. (d) NNCNet. (e) HCT. (f) ExViT. (g) FGCN. (h) MFT. (i) SPT.

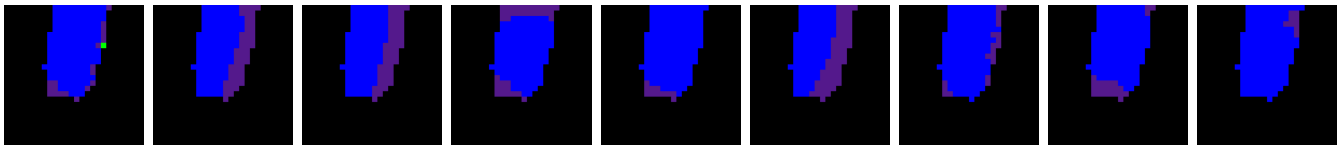


Fig. 8. Classification maps of a local region obtained by different methods on the Trento dataset. (a) SVM. (b) Coupled-CNN. (c) DSHFNet. (d) NNCNet. (e) HCT. (f) ExViT. (g) FGCN. (h) MFT. (i) SPT.

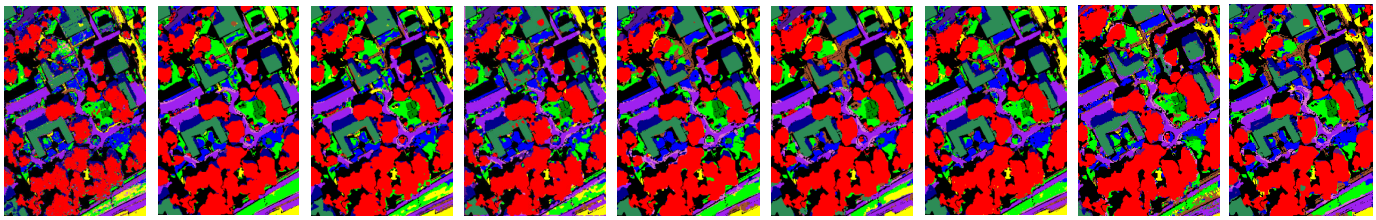
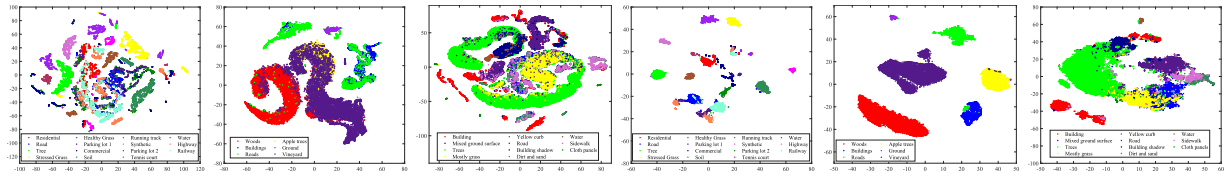


Fig. 9. Classification maps of different methods on the MMUFL dataset. (a) SVM. (b) Coupled-CNN. (c) DSHFNet. (d) NNCNet. (e) HCT. (f) ExViT. (g) FGCN. (h) MFT. (i) SPT.

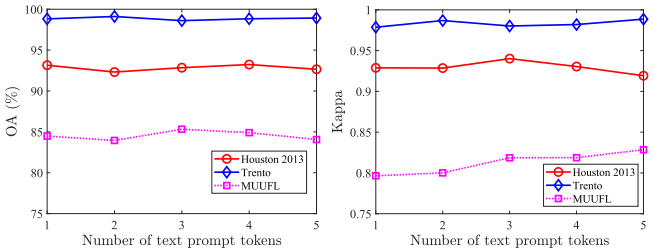
other methods may misclassify Commercial as Trees, Tennis Court, or Soil. In the local region on the Trento dataset, all the methods suffer from misclassification. Among them, the number of pixels misclassified by SPT is relatively small. Besides, SPT can map the original features with severe class mixing phenomenon into a

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(a) (b) (c) (d) (e) (f)

Fig. 10. T-SNE visualization of features. (a) Original features on the Houston 2013 dataset. (b) Original features on the Trento dataset. (c) Original features on the MUUFL dataset. (d) Features extracted by SPT on the Houston 2013 dataset. (e) Features extracted by SPT on the Trento dataset. (f) Features extracted by SPT on the MUUFL dataset.



(a) (b)

Fig. 11. OA and Kappa versus number of TP tokens. (a) OA versus number of TP tokens. (b) Kappa versus number of TP tokens.

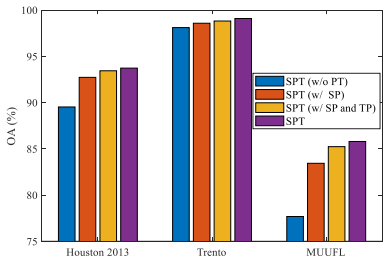
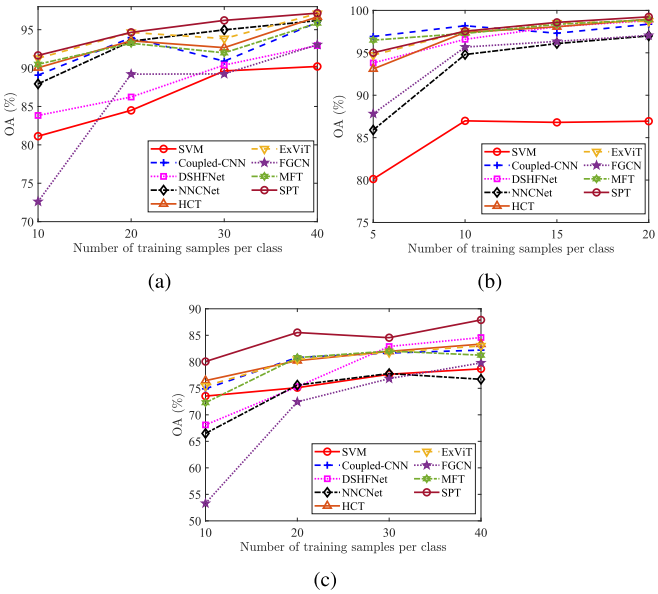


Fig. 12. OA results of ablation study.



(a) (b) (c)

Fig. 13. Relationship between the number of training samples and OA on different datasets. (a) Houston 2013. (b) Trento. (c) MUUFL.

more discriminative feature space. In this feature space, samples of different classes are more distinguishable.

5) The consumed time of DL methods is longer than that of the shallow method (e.g., SVM), which is caused by the large number of parameters and iterative learning process of deep networks. Among all the classifiers, the consumed time of SPT is neither the longest nor the shortest, falling within an acceptable range.

B. Parameter Analysis

In this section, the impact of the number of TP tokens on OA and Kappa will be analyzed. The OAs and Kappa coefficients achieved with different numbers of TP tokens on different datasets are shown in Fig. 11. As the figure illustrates, as the number of TP tokens increases, the OA and Kappa coefficient generally show a trend of first rising and then declining. On one hand, when fewer TP tokens are chosen, it means fewer learnable prompt vectors are available, thereby limiting the representational ability of the class TPs and leading to the problem of insufficient representation of class semantics. On the other hand, when a larger number of TP tokens are chosen, it leads to a larger overall number of parameters to be learned, which in turn results in insufficient optimization of the prompt vectors. This not only increases the risk of overfitting but also causes the proportion of the true class embedding vectors to decrease, thereby leading to a large amount of noise in the class embedding vectors. Therefore, we recommend choosing the number of TP tokens in the range of $\{1, 2, 3, 4, 5\}$. In this article, the number of TP tokens is set 2 on both the MUUFL and Trento datasets, and 1 on the Houston 2013 dataset.

We also conduct ablation experiments to study the influence of different PT methods on OA. The experimental results on the three datasets are shown in Fig. 12, where SPT (w/o PT) means SPT without SPs and TPs. This signifies that only the PE block has been added to the standard CLIP, excluding the SPL block and TPs. SPT (w/ SP) means SPT with SP only, which implies that both the PE and the SPL have been added to the standard CLIP. SPT (w/ SP and TP) means SPT equipped with both SP and TP, indicating that all the three components have been added to the standard CLIP. It can be seen that SPT can achieve the highest OA on all the datasets because it not only uses SP to incorporate abundant spectral information into the process of visual feature extraction but also enhances the discriminative ability of the classification model through TP and CD loss.

Furthermore, we conduct the corresponding experiments to analyze the robustness of the proposed method to the number of training samples. As can be seen from Fig. 13, the OA achieved by the different methods generally shows an increasing trend as the number of training samples increases.

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In a few cases, there is a decrease in OA as the number of training samples increases, and the factor leading to this phenomenon may be the presence of outlier samples, which makes the model overfit to these samples in the case of limited labeled samples. Despite the limited number of labeled samples, the proposed method is still able to achieve competitive performance with the comparison methods.

IV. Conclusion

We attempted to generalize a commonly used VLM, specifically CLIP, to the joint classification task of HSI and LiDAR data. To achieve this and preserve CLIP's posterior vision–text common feature space, we conducted PT from both textual and visual perspectives. By introducing sample-dependent SPs to the vision encoder, CLIP is empowered to extract HSI and LiDAR features containing abundant spectral and elevation information simultaneously, thereby enhancing its ability to recognize ground objects. The inclusion of TPs, coupled with the construction of a CD loss, increases the distance between text features of different classes, thereby enhancing the discriminability of classes with strong semantic similarities. The experimental results on the Houston 2013, Trento, and MUUFL datasets validate the effectiveness of the proposed method. Since multimodal remote sensing data acquired in the real world are often noisy, it is a valuable future direction to obtain more robust multimodal prompts.

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